Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant

Aim:

To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB.

Objectives:

1. Determine the capacitance of an isolated sphere for a specified radius and dielectric constant.

2. Determine the capacitance of a concentric spherical capacitor for specified inner and outer radii and dielectric constant.

3. Investigate the effect of inner radius, outer radius and relative permittivity on capacitance.

4. Compare MATLAB results with the analytical expressions and interpret the observed trends.

Test Case (i) – Capacitance of an Isolated Sphere Determine the capacitance for a specified radius and relative permittivity.

```
clc; clear;

epsilon0 = 8.854e-12;

a = input('Enter radius of isolated sphere (m): ');

er = input('Enter relative permittivity: ');

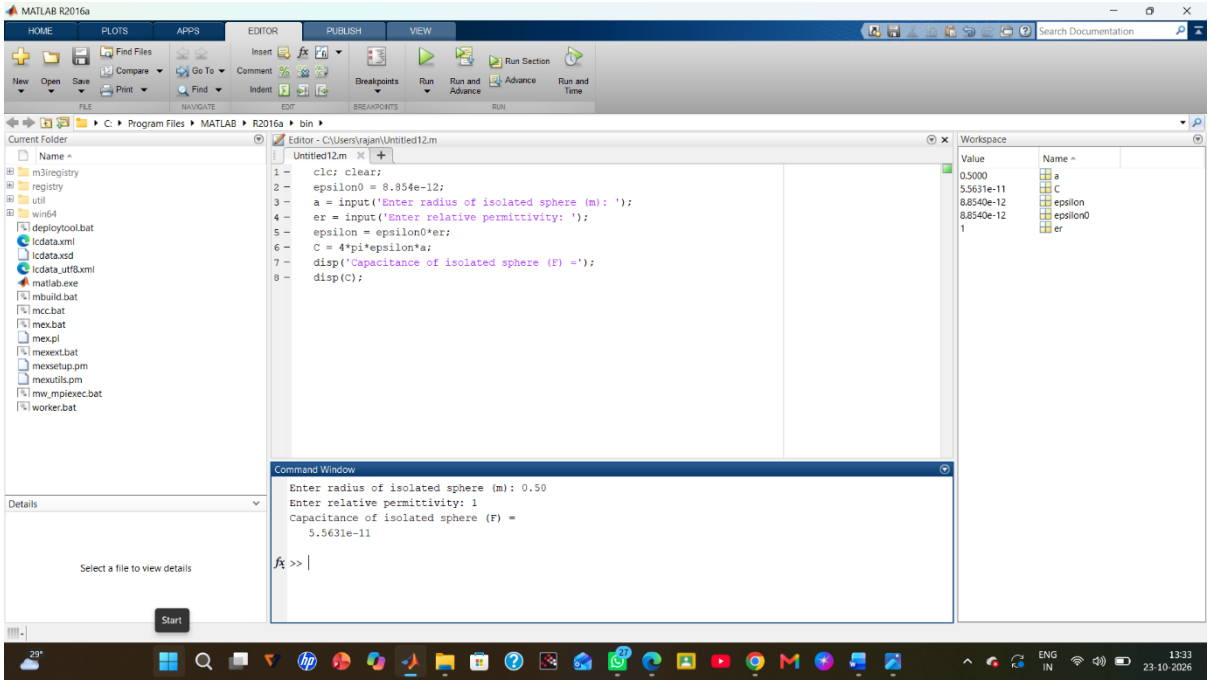
epsilon = epsilon0*er;

C = 4*pi*epsilon*a;

disp('Capacitance of isolated sphere (F) =');

disp(C);
```

Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant



Test Case (ii) – Variation of Isolated-Sphere Capacitance with Radius

```
clc; clear;

epsilon0 = 8.854e-12; er = 1;

a = linspace(0.1,2,100);

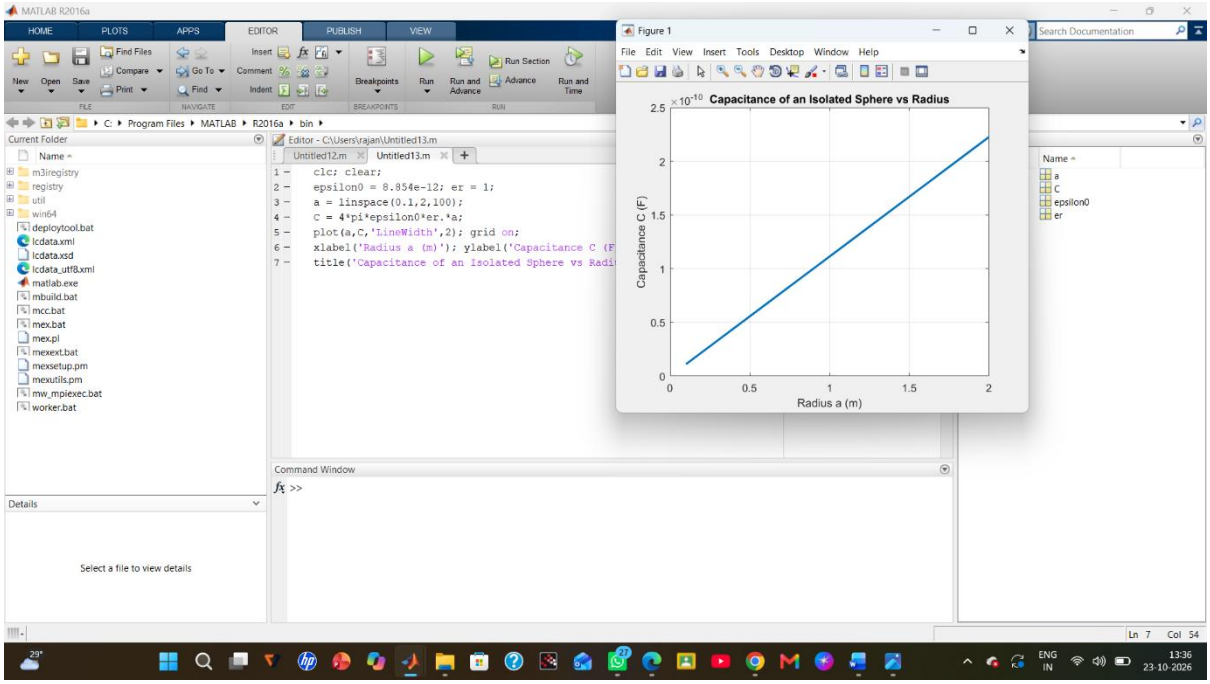
C = 4*pi*epsilon0*er.*a;

plot(a,C,'LineWidth',2); grid on;

xlabel('Radius a (m)'); ylabel('Capacitance C (F)');

title('Capacitance of an Isolated Sphere vs Radius');
```

Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant



Test Case (iii) – Capacitance of a Concentric Spherical Capacitor Determine the capacitance for specified inner radius a, outer radius b and relative permittivity εr.

```
clc; clear;

epsilon0 = 8.854e-12;

a = input('Enter inner radius a (m): ');

b = input('Enter outer radius b (m): ');

er = input('Enter relative permittivity: ');

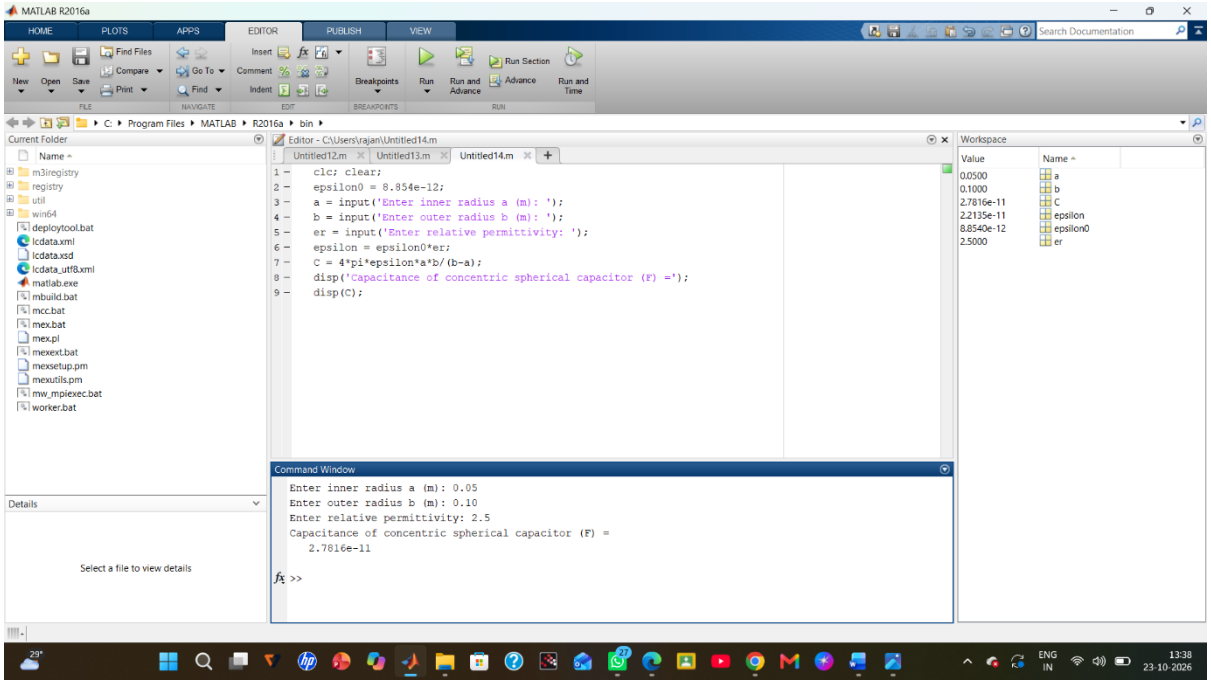
epsilon = epsilon0*er;

C = 4*pi*epsilon*a*b/(b-a);

disp('Capacitance of concentric spherical capacitor (F) =');

disp(C);
```

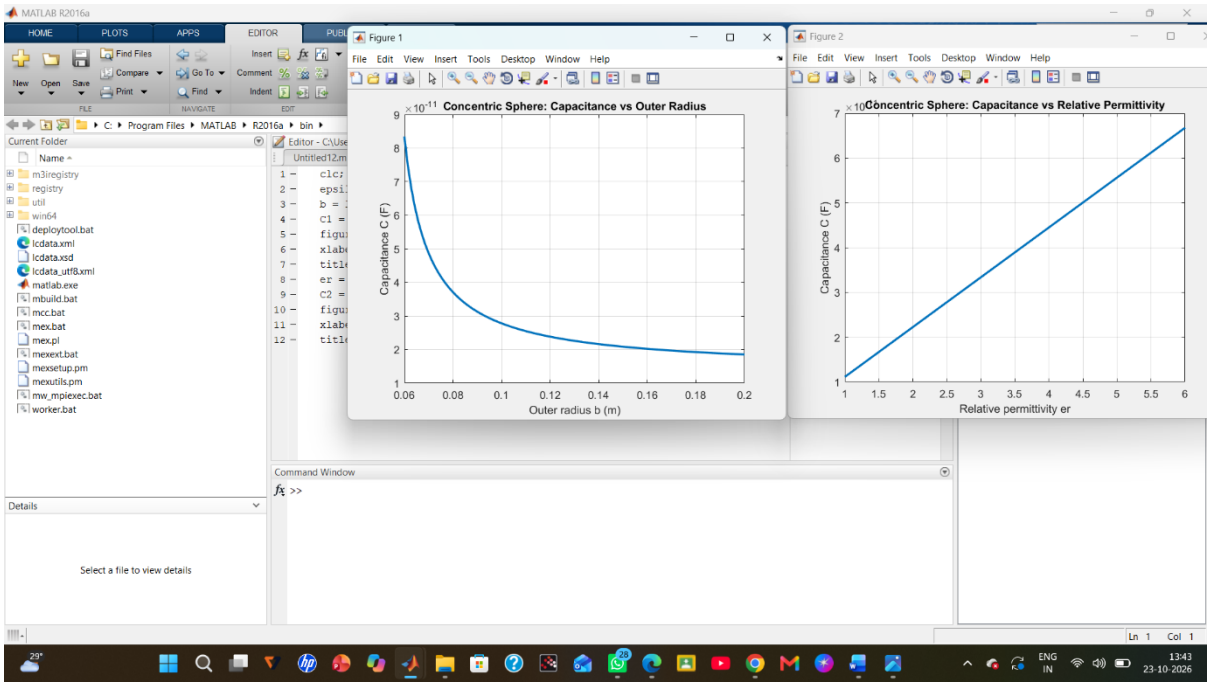
Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant



Test Case (iv) – Effect of Outer Radius and Relative Permittivity

```
clc; clear;
epsilon0 = 8.854e-12; a = 0.05; er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure; plot(b,C1,'LineWidth',2); grid on;
xlabel('Outer radius b (m)'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100); b = 0.10;
C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure; plot(er,C2,'LineWidth',2); grid on;
xlabel('Relative permittivity er'); ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```

Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant



Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)		Marks given
1	Define spherical geometry and dielectric	Define spherical geometry and dielectric	Correctly identifies isolated sphere radius or inner/outer radii and medium permittivity.	Geometr y is correct with minor omission.	Some dimensions or material properties are unclear.	Uses invalid geometry or material inputs.	25		2.5
2	Implement spherical capacitance equations	Implement spherical capacitance equations	Applies the appropriate isolated/con centric sphere formulas correctly	Formulas are mostly correct with minor errors.	Partial calculations require guidance.	Uses incorrect equations.	25		2.5
3	Analyze radius and permittivity effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omission s.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30		3
4	Present results and conclusions	Present results and conclusions	Reports geometry, units, calculations	Report is clear with minor gaps.	Report lacks some key information.	Results are unclear or missing.	20		2

Exp. No: 7 Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant

			and conclusions clearly.						
Total							100		10